

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

(a) Find $\mathbf{M}^{-1}$ giving each element in exact form.

(2)

$$2x + y - 3z = -4$$

$$4x - 2y + z = 9$$

$$3x + 5y - 2z = 5$$

(2)

(1)

5.

$$\mathbf{M} = \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$$

- (a) Show that $\mathbf{M}$ is non-singular.

(2)

The hexagon R is transformed to the hexagon S by the transformation represented by the matrix $\mathbf{M}$.

Given that the area of hexagon R is 5 square units,

- (b) find the area of hexagon S .

(1)

The matrix $\mathbf{M}$ represents an enlargement, with centre $(0, 0)$ and scale factor k , where $k > 0$, followed by a rotation anti-clockwise through an angle θ about $(0, 0)$.

- (c) Find the value of k .

(2)

- (d) Find the value of θ .

(2)

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

1.

$$\begin{pmatrix} x & 9 \\ y & z \end{pmatrix} - 3 \begin{pmatrix} z & y \\ z & y \end{pmatrix} = k\mathbf{I}$$

where x , y , z and k are constants.

Determine the value of x , the value of y and the value of z .

(4)



3.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

- (a) Describe fully the single geometric transformation A represented by the matrix $\mathbf{A}$. (2)

$$\mathbf{B} = \begin{pmatrix} 1 & 3 & 0 \\ \sqrt{3} & 0 & 5\sqrt{3} \\ 1 & 2 & 0 \end{pmatrix}$$

The transformation B is represented by the matrix $\mathbf{B}$.

The transformation A followed by the transformation B is the transformation C , which is represented by the matrix $\mathbf{C}$.

To determine matrix $\mathbf{C}$, a student attempts the following matrix multiplication.

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 1 & 3 & 0 \\ \sqrt{3} & 0 & 5\sqrt{3} \\ 1 & 2 & 0 \end{pmatrix}$$

- (b) State the error made by the student. (1)

- (c) Determine the correct matrix $\mathbf{C}$. (1)



DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA**

DO NOT WRITE IN THIS AREA

- DO NOT WRITE IN THIS AREA**

DO NOT WRITE IN THIS AREA

-

9. (i)
$$\mathbf{P} = \begin{pmatrix} k & -2 & 7 \\ -3 & -5 & 2 \\ k & k & 4 \end{pmatrix} \quad \text{where } k \text{ is a constant}$$

Show that $\mathbf{P}$ is non-singular for all real values of k .

(4)

(ii)
$$\mathbf{Q} = \begin{pmatrix} 2 & -1 \\ -3 & 0 \end{pmatrix}$$

The matrix $\mathbf{Q}$ represents a linear transformation T

Under T , the point $A(a, 2)$ and the point $B(4, -a)$, where a is a constant, are transformed to the points A' and B' respectively.

Given that the distance $A'B'$ is $\sqrt{58}$, determine the possible values of a .

(5)



2. With respect to the **right-hand rule**, a rotation through θ° anticlockwise about the z -axis is represented by the matrix
- $$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Given that the matrix $\mathbf{M}$, where

$$\mathbf{M} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

represents a rotation through α° anticlockwise about the z -axis with respect to the **right-hand rule**,

- (a) determine the value of α . (1)

- (b) Hence determine the smallest possible positive integer value of k for which $\mathbf{M}^k = \mathbf{I}$ (2)

The 3×3 matrix $\mathbf{N}$ represents a reflection in the plane with equation $y = 0$

- (c) Write down the matrix $\mathbf{N}$. (1)

The point A has coordinates $(-2, 4, 3)$

The point B is the image of the point A under the transformation represented by matrix $\mathbf{M}$ followed by the transformation represented by matrix $\mathbf{N}$.

- (d) Show that the coordinates of B are $(2 + \sqrt{3}, 2\sqrt{3} - 1, 3)$ (2)

Given that O is the origin,

- (e) show that, to 3 significant figures, the size of angle AOB is 66.9° (2)

- (f) Hence determine the area of triangle AOB , giving your answer to 3 significant figures. (2)



3. With respect to the **right-hand rule**, a rotation through θ° anticlockwise about the y -axis is represented by the matrix

$$\begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

The point P has coordinates $(8, 3, 2)$

The point Q is the image of P under the transformation reflection in the plane $y = 0$

- (a) Write down the coordinates of Q (1)

The point R is the image of P under the transformation rotation through 120° anticlockwise about the y -axis, with respect to the **right-hand rule**.

- (b) Determine the exact coordinates of R (2)

- (c) Hence find $|\vec{PR}|$ giving your answer as a simplified surd. (2)

- (d) Show that $\overrightarrow{PR}$ and $\overrightarrow{PQ}$ are perpendicular. (1)

- (e) Hence determine the exact area of triangle PQR , giving your answer as a surd in simplest form. (2)



6. (i)

$$\mathbf{A} = \begin{pmatrix} 2 & a \\ a-4 & b \end{pmatrix}$$

where a and b are non-zero constants.

Given that the matrix $\mathbf{A}$ is self-inverse,

- (a) determine the value of b and the possible values for a . (5)

The matrix $\mathbf{A}$ represents a linear transformation M .

Using the smaller value of a from part (a),

- (b) show that the invariant points of the linear transformation M form a line, stating the equation of this line. (3)

(ii)

$$\mathbf{P} = \begin{pmatrix} p & 2p \\ -1 & 3p \end{pmatrix}$$

where p is a positive constant.

The matrix $\mathbf{P}$ represents a linear transformation U .

The triangle T has vertices at the points with coordinates $(1, 2)$, $(3, 2)$ and $(2, 5)$.

The area of the image of T under the linear transformation U is 15

- (a) Determine the value of p . (4)

The transformation V consists of a stretch scale factor 3 parallel to the x -axis with the y -axis invariant followed by a stretch scale factor -2 parallel to the y -axis with the x -axis invariant. The transformation V is represented by the matrix $\mathbf{Q}$.

- (b) Write down the matrix $\mathbf{Q}$. (2)

Given that U followed by V is the transformation W , which is represented by the matrix $\mathbf{R}$,

- (c) find the matrix $\mathbf{R}$. (2)



3. Tyler invested a total of £5000 across three different accounts; a savings account, a property bond account and a share dealing account.

Tyler invested £400 more in the property bond account than in the savings account.

After one year

- the savings account had increased in value by 1.5%
- the property bond account had increased in value by 3.5%
- the share dealing account had **decreased** in value by 2.5%
- the total value across Tyler's three accounts had increased by £79

Form and solve a matrix equation to find out how much money was invested by Tyler in each account.

(7)

1.

$$\mathbf{A} = \begin{pmatrix} 4 & -1 \\ 7 & 2 \\ -5 & 8 \end{pmatrix} \quad \mathbf{B} = \begin{pmatrix} 2 & 3 & 2 \\ -1 & 6 & 5 \end{pmatrix} \quad \mathbf{C} = \begin{pmatrix} -5 & 2 & 1 \\ 4 & 3 & 8 \\ -6 & 11 & 2 \end{pmatrix}$$

Given that $\mathbf{I}$ is the 3×3 identity matrix,

- (a) (i) show that there is an integer k for which

$$\mathbf{AB} - 3\mathbf{C} + k\mathbf{I} = \mathbf{0}$$

stating the value of k

- (ii) explain why there can be no constant m such that

$$\mathbf{BA} - 3\mathbf{C} + m\mathbf{I} = \mathbf{0}$$

(4)

- (b) (i) Show how the matrix $\mathbf{C}$ can be used to solve the simultaneous equations

$$-5x + 2y + z = -14$$

$$4x + 3y + 8z = 3$$

$$-6x + 11y + 2z = 7$$

- (ii) Hence use your calculator to solve these equations.

(3)



1. A system of three equations is defined by

$$\begin{aligned}kx + 3y - z &= 3 \\ 3x - y + z &= -k \\ -16x - ky - kz &= k\end{aligned}$$

where k is a positive constant.

Given that there is no unique solution to all three equations,

- (a) show that $k = 2$

(2)

Using $k = 2$

- (b) determine whether the three equations are consistent, justifying your answer.

(3)

- (c) Interpret the answer to part (b) geometrically.

(1)



10. The population of chimpanzees in a particular country consists of juveniles and adults. Juvenile chimpanzees do not reproduce.

In a study, the numbers of juvenile and adult chimpanzees were estimated at the start of each year. A model for the population satisfies the matrix system

$$\begin{pmatrix} J_{n+1} \\ A_{n+1} \end{pmatrix} = \begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix} \begin{pmatrix} J_n \\ A_n \end{pmatrix} \quad n = 0, 1, 2, \dots$$

where a is a constant, and J_n and A_n are the respective numbers of juvenile and adult chimpanzees n years after the start of the study.

- (a) Interpret the meaning of the constant a in the context of the model.

(1)

At the start of the study, the total number of chimpanzees in the country was estimated to be 64 000

According to the model, after one year the number of juvenile chimpanzees is 15 360 and the number of adult chimpanzees is 43 008

- (b) (i) Find, in terms of a

$$\begin{pmatrix} a & 0.15 \\ 0.08 & 0.82 \end{pmatrix}^{-1}$$

(3)

- (ii) Hence, or otherwise, find the value of a .

(3)

- (iii) Calculate the change in the number of juvenile chimpanzees in the first year of the study, according to this model.

(2)

Given that the number of juvenile chimpanzees is known to be in decline in the country,

- (c) comment on the short-term suitability of this model.

(1)

A study of the population revealed that adult chimpanzees stop reproducing at the age of 40 years.

- (d) Refine the matrix system for the model to reflect this information, giving a reason for your answer.

(There is no need to estimate any unknown values for the refined model, but any known values should be made clear.)

(2)

